CSE 242 – Assignment 3

Yi Zhong
yzhong76@ucsc.edu

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1 Hard-margin linear SVM

Positive points: $1 = (1, 1)$, $2 = (1, 2)$, $3 = (2, 1)$; negative points: $4 = (0, 0)$, $5 = (1, 0)$, $6 = (0, 1)$.

(a)

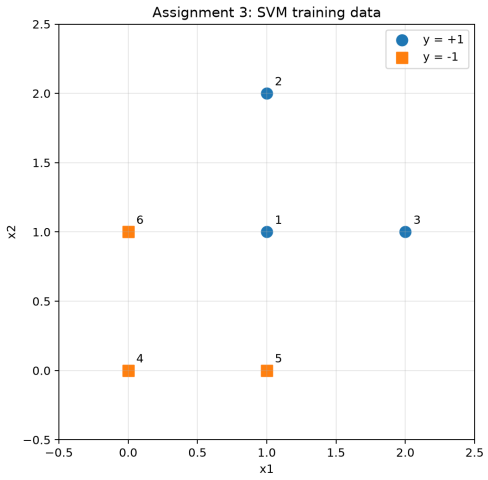


Figure 1: Training data with point IDs.

Support vectors: points **1, 5, 6**. Point 1 lies on $x_1 + x_2 = 2$ and points 5, 6 lie on $x_1 + x_2 = 1$; these are the closest points of opposite classes, so the widest gap is between these two lines and the boundary is halfway between them. Points 2, 3 ($x_1 + x_2 = 3$) and 4 ($x_1 + x_2 = 0$) are farther from the gap.

(b) Each support vector x_i with label y_i lies on its supporting line, $y_i(w_1 x_{i1} + w_2 x_{i2} + b) = 1$:

$$\begin{aligned}
&\text{point 1, } y = +1: \quad w_1 + w_2 + b = 1 \\
&\text{point 5, } y = -1: \quad -(w_1 + b) = 1 \;\Rightarrow\; w_1 + b = -1 \\
&\text{point 6, } y = -1: \quad -(w_2 + b) = 1 \;\Rightarrow\; w_2 + b = -1
\end{aligned}$$

Subtracting the second equation from the first gives $w_2 = 2$, and subtracting the third from the first gives $w_1 = 2$; then $b = -1 - w_1 = -3$:

$$w = (2, 2), \quad b = -3, \qquad \text{boundary } 2x_1 + 2x_2 - 3 = 0 \;\; (x_1 + x_2 = 1.5).$$

Geometric margin (distance from the boundary to the closest points), with $\|w\| = \sqrt{2^2 + 2^2} = 2\sqrt{2}$:

$$\gamma = \frac{|w \cdot x_1 + b|}{\|w\|} = \frac{|2 + 2 - 3|}{2\sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4} \approx 0.354.$$

2 K-means

(a) Ties: a point at equal distance from several centroids goes to the one with the smallest index. Empty clusters: an empty cluster keeps its previous centroid.

(b)

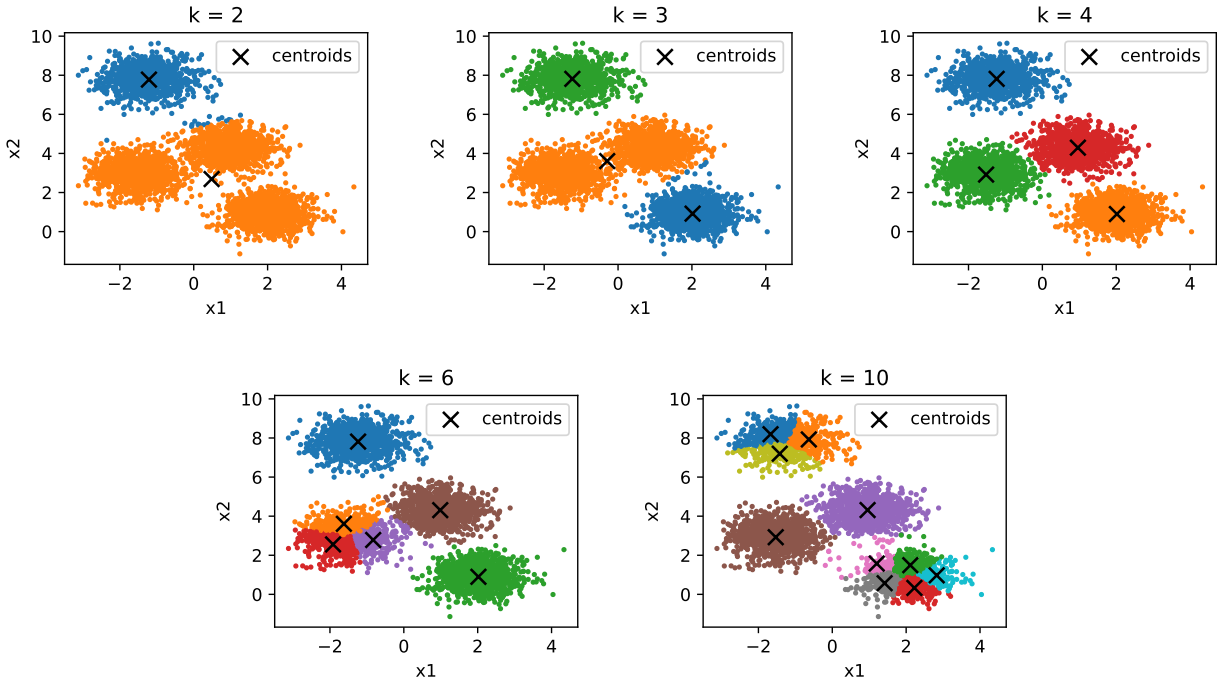


Figure 2: K-means cluster assignments and final centroids for $k = 2, 3, 4, 6, 10$.

With $k = 2$ the top group around $(-1.2, 7.8)$ forms one cluster and the other three groups are merged into the second. With $k = 3$ the group around $(2.0, 0.9)$ separates and the two middle groups around $(-1.5, 2.9)$ and $(1.0, 4.3)$ remain merged. With $k = 4$ each centroid lands in one generating group, so the clusters match the four groups. With $k = 6$ and $k = 10$ the extra centroids split single groups into pieces, so the clusters become smaller and tighter but no longer correspond to the generating groups.

3 Noisy-label SVM and K-NN repair

(a) Mean clean-test accuracy over 20 trials (seed 0): **0.8280**.

(b)

Method	Mean test accuracy (20 trials)
Baseline (noisy labels)	0.8280
K-NN repaired labels	0.8644
Difference (K-NN − baseline)	+0.0364

K-NN repair helped. Positive labels are flipped more often (0.35 vs. 0.20), so the noisy training set has too few positive labels and the baseline SVM predicts -1 over part of the positive region. The positive class is compact (covariance I), so most neighbors of a positive point still have label $+1$ and the vote flips its label back. Repair can hurt where most neighbors are mislabeled, such as the overlap of the two classes and the sparse tails of the negative class; there the vote replaces a correct label with a wrong one.